W13 · EMNLP 2026 WORKSHOPS
ArabicNLP 2026 · Shared Tasks
The Fourth Arabic Natural Language Processing Conference

› ALEXIS AT ALEXANDRIAX-2026 SUBTASK 2

SHARED TASK AlexandriaX 2026 SUBTASK 2

Capacity Discovery and Calibrated Preference Optimization for Cross-Dialect Arabic MT

2nd OF 4 TEAMS

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25.84 Official spBLEU — rank 2 / 4 · chrF++ 39.02
28.04 Best post-evaluation — DPO Config B · 0.36 from 1st
+8.9% One rank + epoch step — beat three data rounds combined
70,139 Training pairs — 3 trust tiers · 38 directions

01 › THE TASK

Translate between MSA and six dialects with almost no parallel text

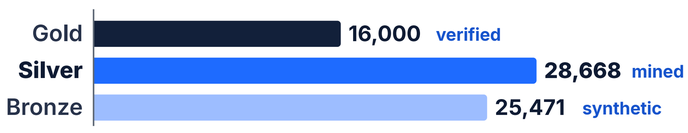
Each test instance is a (source dialect, sentence, target dialect) triple over short **banking-support** sentences. MSA appears only as a source or pivot, never as a scored target. A blind test set of **3,574 rows** is scored with **spBLEU** (FLORES-200 SentencePiece) and **chrF++**.

Egyptian no parallel text	**Lebanese** no parallel text	**Moroccan** mined pairs
Palestinian gold + mined	**Saudi** mined pairs	**Tunisian** mined pairs

Short, formulaic sentences: the obstacle is uneven data scarcity. Egyptian + Lebanese ≈ 39% of test rows (our estimate, not an official breakdown).

02 › TRAINING DATA

A three-tier corpus built from ArBanking77



Gold VERIFIED	Real, human-written MSA↔Palestinian pairs; the only tier with a genuine ground-truth target.
Silver TOPIC-MATCHED	Cross-dialect pairs mined by embedding similarity inside a shared banking-intent bucket (MSA, Pal., Mor., Sau., Tun.): same topic, not a verified translation.
Bronze BACK-TRANSLATED	Synthetic Egyptian and Lebanese pairs, back-translated through the model itself and filtered for degenerate output.

Backbone for every experiment: **facebook/nllb-200-distilled-600M**, adapted first with LoRA, then by full fine-tuning.

03 › SILENT FAILURES WE CAUGHT

Bugs that raise no error

- **NLLB tokenizer:** with `src_lang` the fast tokenizer drops the FLORES-200 source-language token (otherwise it appends a spurious one); we build `[lang_id, content, eos]` by hand.
- **Tunisian artifact:** 29.4% of raw rows carried a stray leading underscore; the model reproduced it in 35% of its Tunisian predictions.
- **peft merge-then-finetune:** merged weights stay frozen; our first full fine-tune trained nothing for two epochs (checkpoint byte-identical to its initialization). Fix: `requires_grad_(True)` + a trainable-parameter assertion.
- **bf16 DPO at lr 1e-5:** updates can fall below the precision floor, a silent no-op. All DPO runs use fp32.

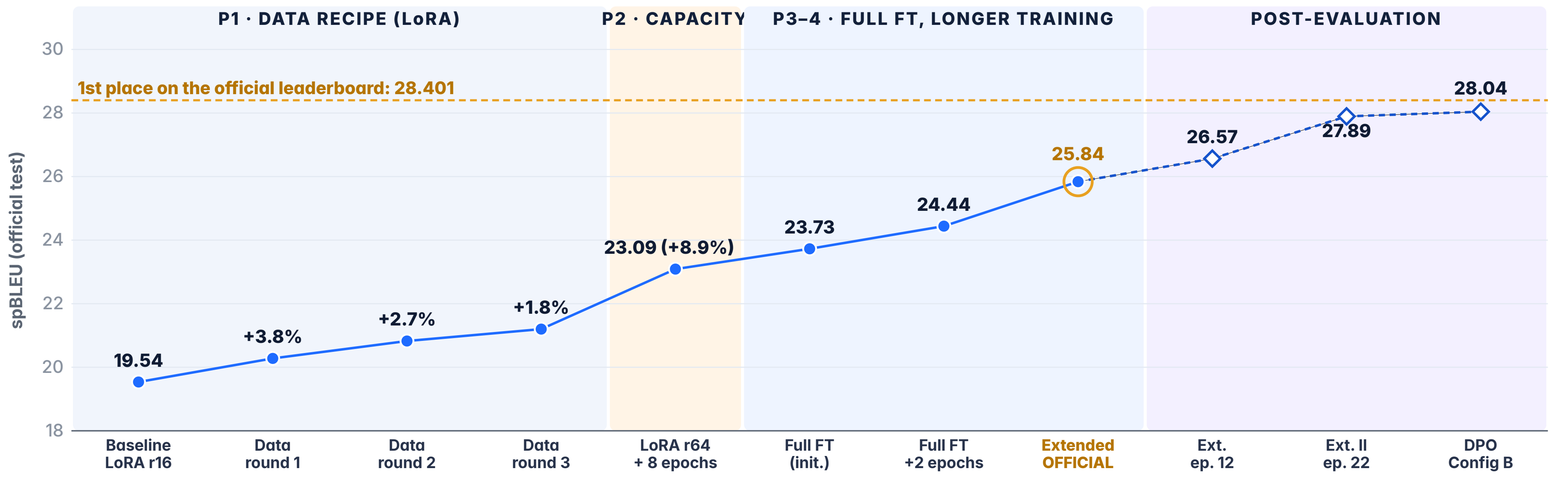
04 › TRIED AND REJECTED

Clean negative results

- MSA-pivot decoding lost to direct decoding in 11 of 12 non-MSA directions.
- Diverse-beam MBR helped 8 of 20 directions (noise level); a decode sweep gained ≈0.02 spBLEU.
- Cross-model consensus reranking failed (the weaker model was wrong more often); checkpoint averaging and greedy model soups never beat the single best checkpoint.
- Exact-match retrieval: 39% of test sources match a training source verbatim, but none from the gold tier, so substitution risked harm.

05 › THE PROJECT ON THE REAL LEADERBOARD

Every configuration scored on Codabench, not on an offline proxy



Real Codabench spBLEU at key milestones. Filled circles: submission phase; open diamonds: post-evaluation (after the deadline, not eligible for ranking). Data-round points are reconstructed from the reported relative gains (+3.8%, +2.7%, +1.8% on 19.536); the paper gives no exact values for them. Official result **25.840** (chrF++ 39.019): 2.56 behind 1st, 1.31 ahead of 3rd. Best post-evaluation checkpoint **28.042** (chrF++ 40.774) closes the gap to 1st to 0.36.

06 › FOUR PHASES

Change the variable when returns flatten

P1 Data-recipe tuning (LoRA)
From a LoRA baseline: widen silver/bronze caps, grow gold 2,500 → 16,000, fix the Tunisian artifact. Every round helped less: **+3.8%, +2.7%, +1.8%** relative spBLEU.

P2 Capacity discovery
Data recipe held fixed; LoRA rank 32 → 64 and epoch budget 5 → 8 raised together: **+8.9%**, more than three data rounds combined. Not ablated separately, so it shows the data recipe had plateaued, not capacity alone.

P3 Full fine-tuning
Merge the best adapter, train all weights. Self-distilled bronze scored better offline and **worse on the leaderboard**: no dev set covers Egyptian or Lebanese.

P4 Training duration as a lever
Validation loss was still falling when budgets expired: two extra epochs gave **+0.71** spBLEU, then open-ended rounds of up to 10 epochs with early stopping gave the project's largest cumulative gain.

07 › SUBMISSION HISTORY

Key milestones (of 24 real submissions)

SYSTEM ITERATION	SPBLEU	CHRF++
Baseline LoRA (r = 16)	19.536	35.010
Expanded LoRA (r = 64)	23.088	37.440
Full fine-tune (init.)	23.727	37.730
Full fine-tune (+2 epochs)	24.440	38.259
FT, extended (official)	**25.840**	**39.019**
FT, extended (ep. 12) †	26.565	39.610
FT, extended II (ep. 22) †	27.894	40.538
DPO Config A †	25.017	39.136
DPO Config B (best) †	**28.042**	**40.774**

† post-evaluation phase, scored on Codabench after the official deadline.

08 › PREFERENCE OPTIMIZATION

DPO helps only with the right trust boundary

Sample K = 4 candidates plus the training reference; the best and worst become chosen / rejected. A and B share a Phase-4 checkpoint and differ in reward and β; C continues from B.

A Uniform reward, β = 0.1 — 25.02 REAL SPBLEU, pref. acc. 0.683
chrF++ to the reference + marker bonus for all six dialects. Trusts self-distilled bronze; a real reference is in the pool 99% of the time and usually wins, so DPO mostly re-teaches cross-entropy.

B Trust-restricted reward, β = 0.3 — 28.04 REAL SPBLEU · BEST, pref. acc. 0.691
No bronze references where gold/silver exist; Egyptian and Lebanese get a **reference-free** reward (BGE-M3 similarity to the source + marker bonus).

C B repeated on its own output — not submitted, pref. acc. 0.717
Regressed on every offline metric (silver_dev −5.6), so it was withheld.

TRAINING DYNAMICS DO NOT PREDICT QUALITY
Preference accuracy climbed smoothly for A, B and C alike: it measures how well a policy satisfies its own reward, not real translation quality. **Reward auditing is not optional.**

DO THE OBJECTIVES COMPOUND?
Continuing plain cross-entropy fine-tuning from B scored **27.51**, below 28.04; we read this as plain fine-tuning partly undoing the targeted DPO fix.

09 › SETUP · LIMITATIONS

Key hyperparameters

STAGE	SETTING
LoRA	r ∈ {16, 32, 64} · ≤ 8 ep · lr 1e-4
Full FT	eff. batch 32 · lr 3e-5 · bf16
DPO	lr 1e-5 · β ∈ {0.1, 0.3} · 7 ep · fp32
Decoding	beam 4 · no-repeat 3-gram · rep. 1.3

Limitations: offline metrics directionally, not proportionally, reliable; no seed variance; selection used 24 leaderboard submissions; aggregate scores only. **Next:** a learned reference-free QE reward; the capacity lever at a larger backbone.

TAKE-HOME MESSAGE

When a data recipe's returns decelerate, **test model capacity as an independent lever** before assuming more data tuning is the answer. Preference optimization for MT is only as good as **the trust boundary drawn around its reward's reference data**: training dynamics alone cannot reveal it, and success does not automatically compound under further iteration.

PAPER
ACL Anthology 2026.arabicnlp-sharedtasks.104

CODE
github.com/Momen-Ibrahim-Fawzy/ALEXIS-AlexandriaX-Subtask2-CrossDialectMT

KEY REFERENCES
El Mekki et al. (2026). AlexandriaX 2026: The First Shared Task on Dialectal Arabic Machine Translation. ArabicNLP. · NLLB Team (2022). No Language Left Behind. · Jarrar et al. (2023) ArBanking77 · Hu et al. (2021) LoRA · Rafailov et al. (2023) DPO · Chen et al. (2024) BGE M3 · Wortsman et al. (2022) Model soups · Popović (2017) chrF++ · Post (2018) sacreBLEU.

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